

SiteWhere components include the following:

Core platform: Apache Tomcat 7 (Server Core), a Spring framework (a core configuration framework that performs bootstrapping), Spring Security (the security infrastructure), and Hazelcast (for high performance data processing)

Data storage technologies: MongoDB, Apache HBase and InfluxDB

Integration technologies: Mule AnyPoint platform, Apache Spark

Asset management systems: WSO2 identity server

Features

- Supports the unlimited running of IoT applications on a single instance.
- Devices can be added using self-registration, REST services or batches.
- Controls unlimited devices using batch command operations.
- Integrates with Azure EventHub for massive cloud processing, connects to Twilio for cloud communication, and supports platforms like Arduino and Android.

Official website: <http://www.sitewhere.org/>

DeviceHive

DeviceHive is an open source IoT data platform that facilitates the connectivity of devices to the cloud within a short interval of time, i.e., just a few minutes to allow streaming of data and command traversal. The platform is distributed under the Apache 2.0 licence and is highly scalable via containerisation. It supports Docker Compose and Kubernetes deployment through which public, private or even hybrid clouds can be deployed from a single machine

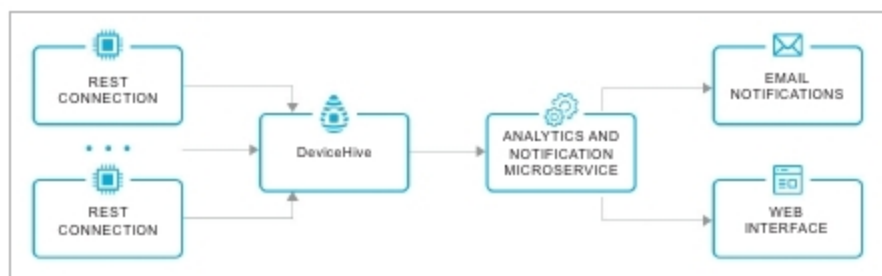


Figure 3: DeviceHive IoT platform

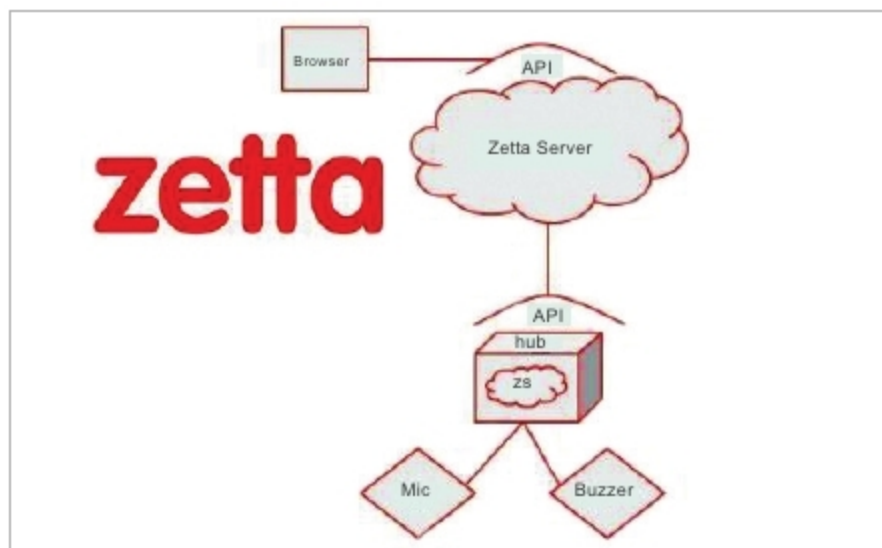


Figure 4: Zetta IoT platform

to extensive enterprise clusters. DeviceHive facilitates connectivity to any device or development board via the REST API, WebSockets or MQTT.

Features

- Highly customised behaviour using JavaScript.
- Supports Big Data solutions like Spark, Cassandra, Kafka and ElasticSearch.
- Allows batch analytics and machine learning on top of device data.
- Connects any device using the REST API, MQTT or WebSockets.
- Direct integration with Alexa.

Official website: <https://devicehive.com/>

Zetta

Zetta is an open source, API based IoT platform built on Node.js. It is regarded as a complete toolkit to generate HTTP APIs for devices. Zetta combines REST APIs, WebSockets and reactive programming to build data-intensive and real-time applications.

Architecture

The Zetta IoT platform's architecture comprises the following components.

- **Zetta Server:** This is the top level of the abstraction component of Zetta. It contains drivers, scouts, apps and server extensions. It runs on hardware like Raspberry Pi, Arduino, BeagleBone, etc.
- **Scouts:** These provide a discovery mechanism for all devices on the network or those that require system resources to speak to a specific protocol. They are also used to identify information for devices, such as a MAC address.
- **Drivers:** These are responsible for modelling devices and interacting with them at a physical level.
- **Server extensions:** These deal with API management, defining additional APIs and adding security to them.
- **Registry:** This is a small database in the context of the server, which stores all the information with regard to devices.

Features

- Can run on the cloud, on a PC or even small development boards.
- Turns any device into an API.
- Easy interface and simple programming to control sensors, actuators and controllers.
- Optimised for data-intensive and real-time applications.
- Allows developers to assemble smartphone apps, device apps and cloud apps together into large, complex adaptive 'systems of systems' to operate in the world of automation.

Official website: <http://www.zettajs.org/>